Assignment Completion Record

Task: Notification System, Email Integration & Real-Time Updates

Project: Smart E-Commerce Platform

Status: ✅ **COMPLETED**

1. Notification System

The complete notification management module has been successfully implemented.

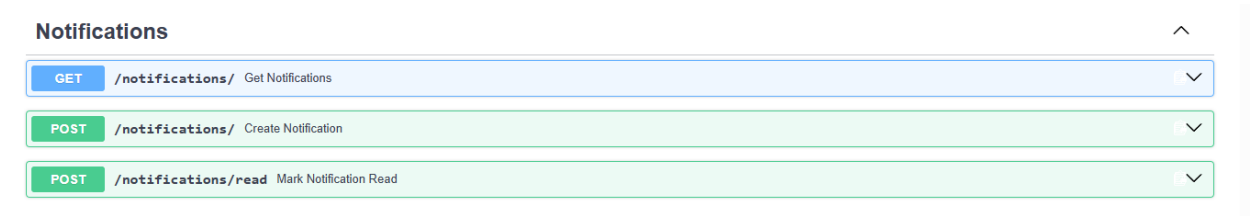
Notification Features Implemented

- Notification creation
- Notification retrieval
- Read/Unread status tracking
- User-specific notifications
- Timestamp management
- Order event notifications
- Payment event notifications
- Shipping notifications
- Delivery notifications

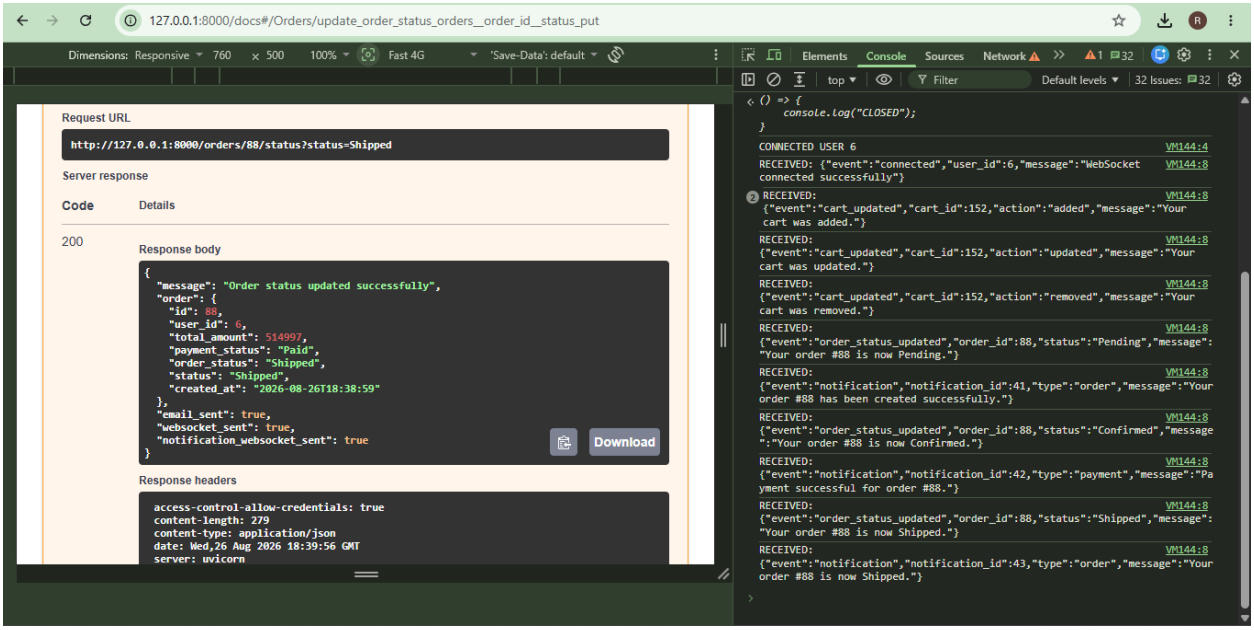
Notification Types

- Order Confirmed
- Payment Successful
- Payment Failed
- Order Shipped
- Order Delivered

1.1 Fig for FastAPI Backend Update for Notifications



1.2 Fig for Notification API Testing (Swagger/Postman)

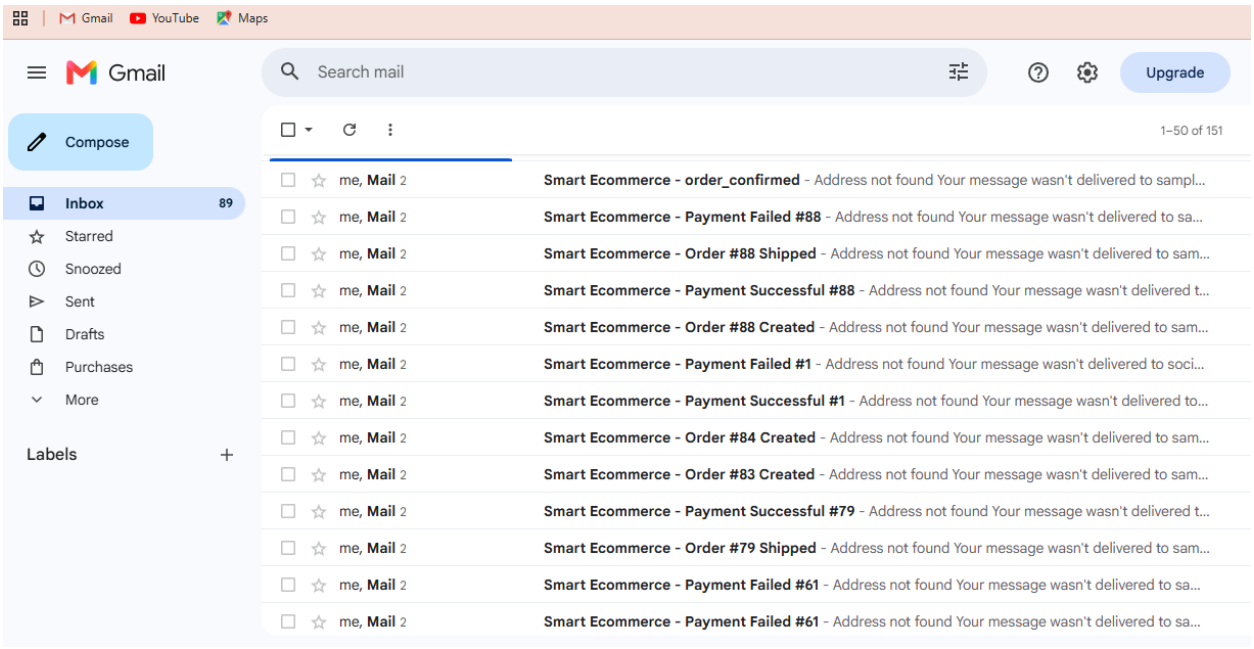


2. Email Notification Integration

The email notification system has been implemented successfully.

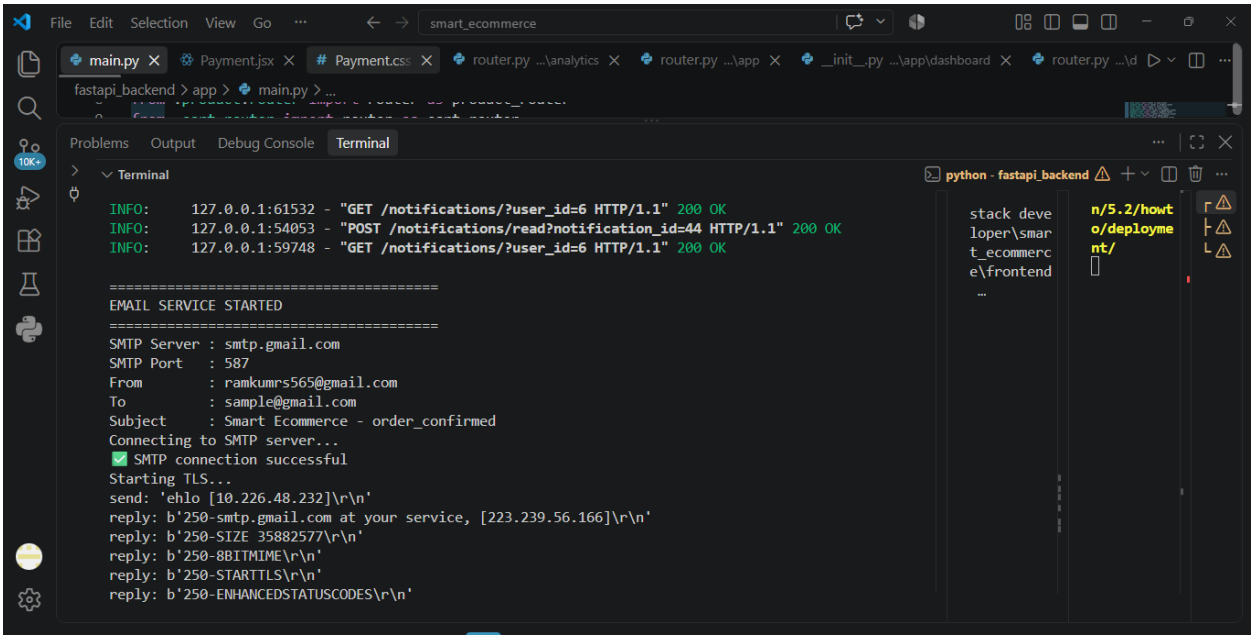
Email Events Implemented

- Order Confirmation Email
- Payment Successful Email
- Payment Failed Email
- Order Shipped Email
- Order Delivered Email



Supported Email Providers

- SMTP
- SendGrid and AWS SES (Ready for Integration)



Email Features

- Dynamic email content
- Event-based email triggers
- Secure email configuration using environment variables
- HTML email support
- User-specific email delivery

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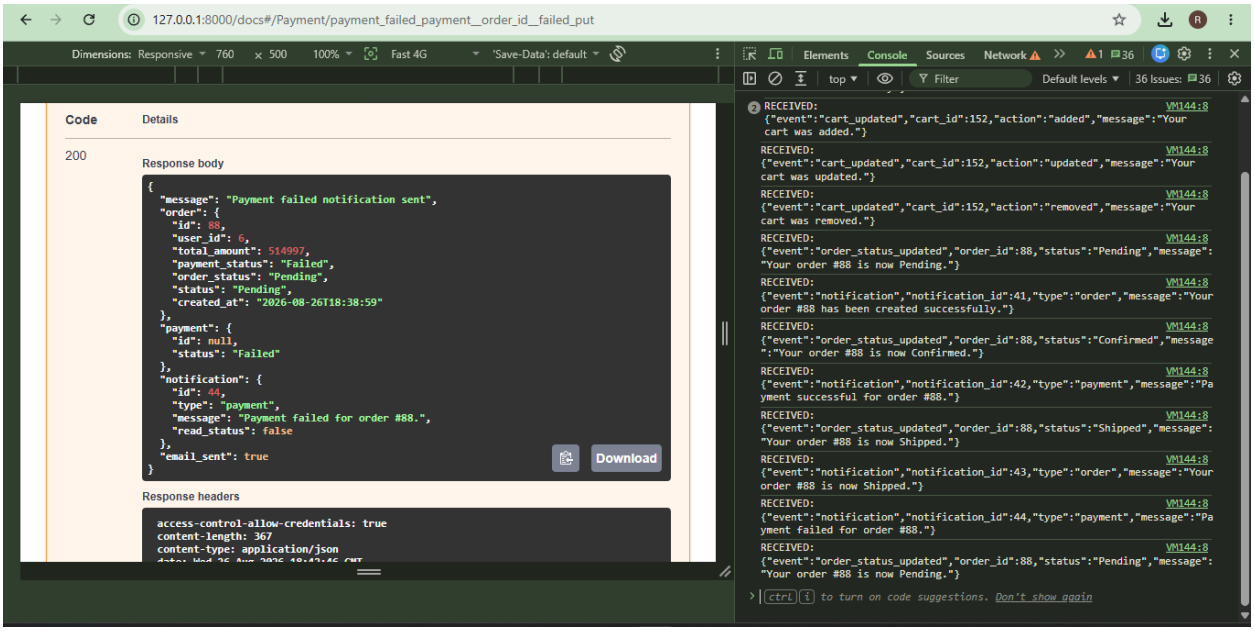


3. Real-Time Notification System

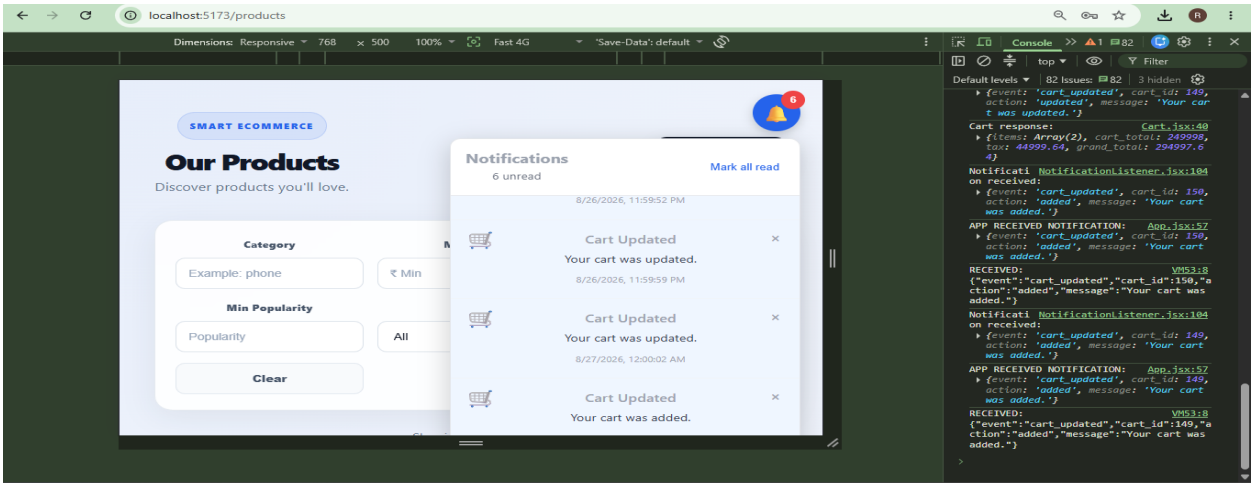
The real-time notification module has been implemented using WebSockets.

Real-Time Features Implemented

- Live notification delivery
- Real-time order updates
- Real-time cart updates
- Instant frontend updates
- Persistent WebSocket connection

WebSocket Events

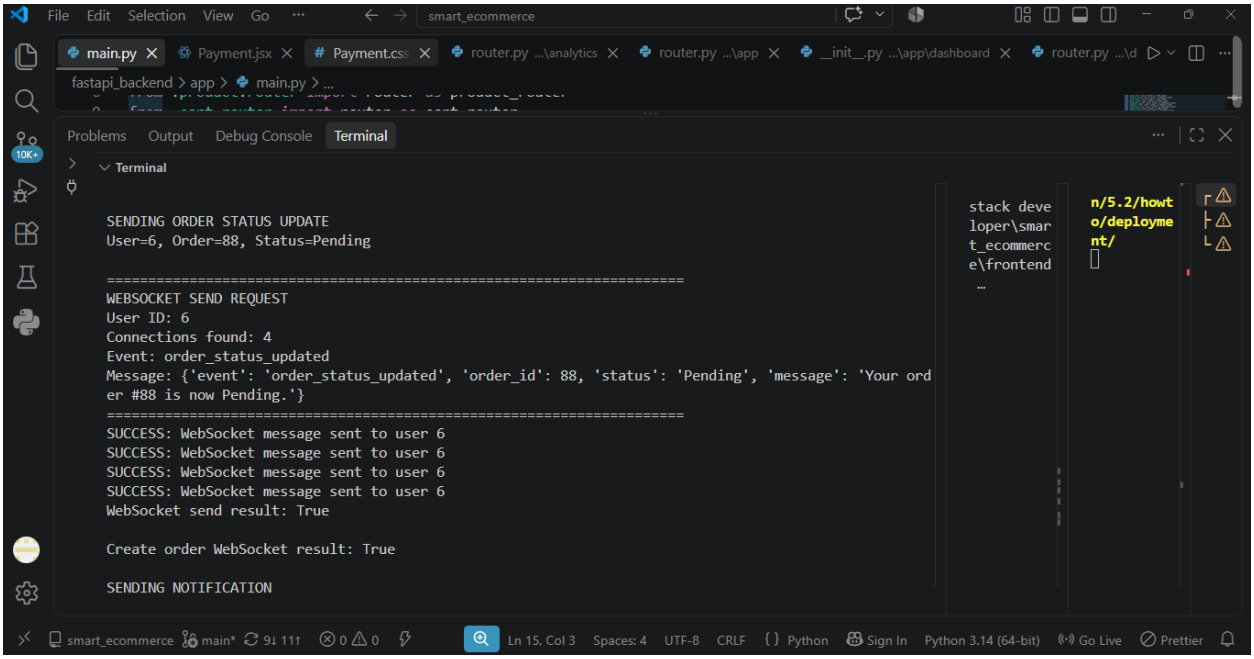
- order_status_updated
- cart_updated



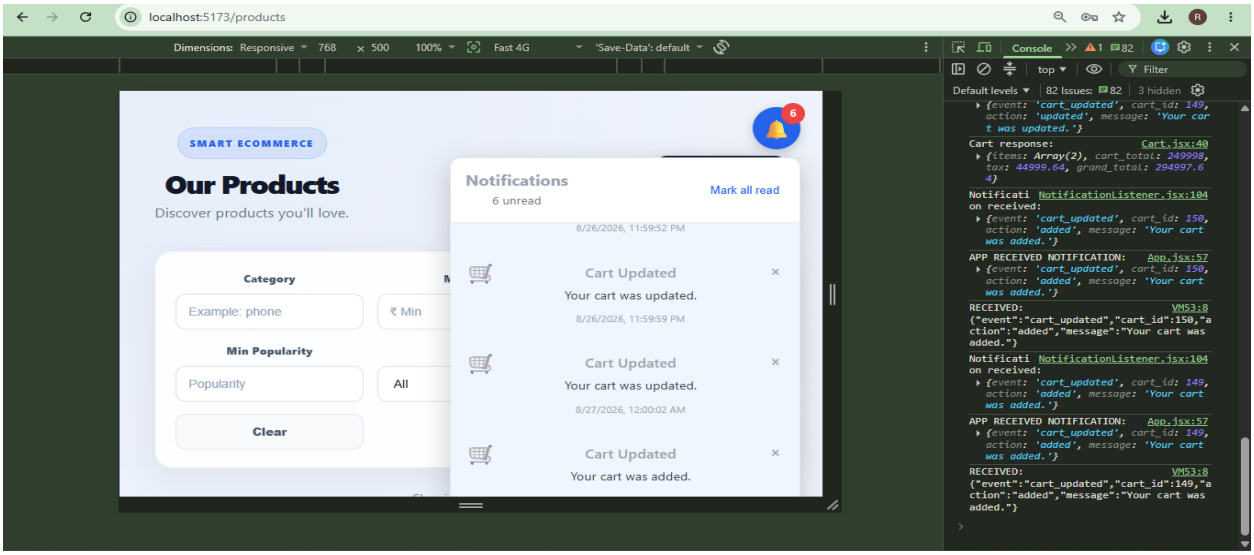
Real-Time Workflow

```
User Action
     ↓
FastAPI Backend
     ↓
Database Update
     ↓
WebSocket Event
     ↓
Connected Client
     ↓
Instant UI Update
```

3.1 Fig for WebSocket Backend Update



3.2 Fig for Real-Time Update



4. Notification APIs

The required Notification APIs have been implemented and tested successfully.

Implemented APIs

Get Notifications

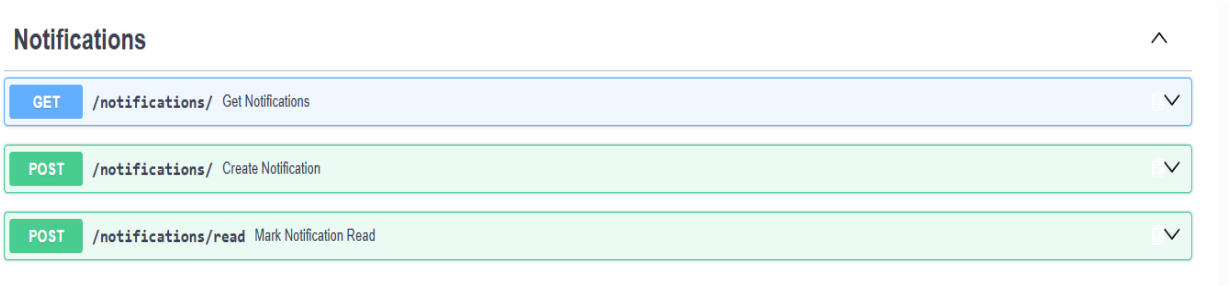
```
GET /notifications
```

Mark Notification as Read

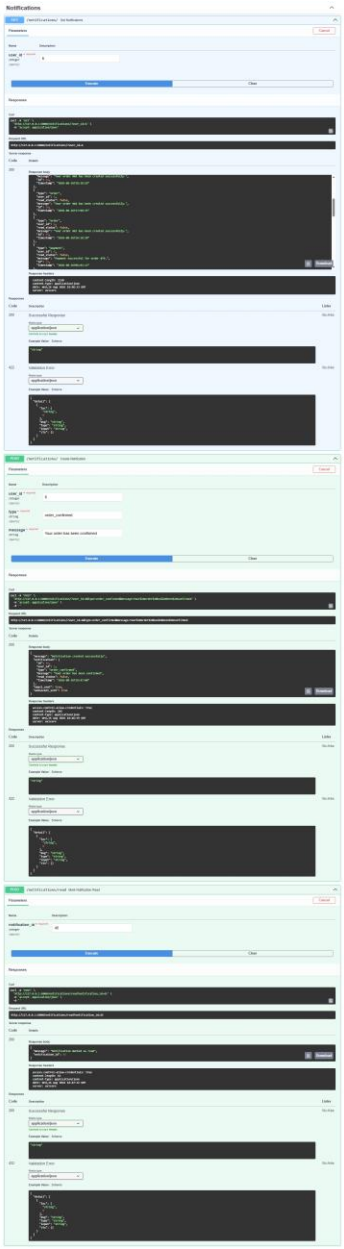
```
POST /notifications/read
```

API Features

- User notification retrieval
- Read status update
- Notification filtering
- Secure access control
- Validation handling

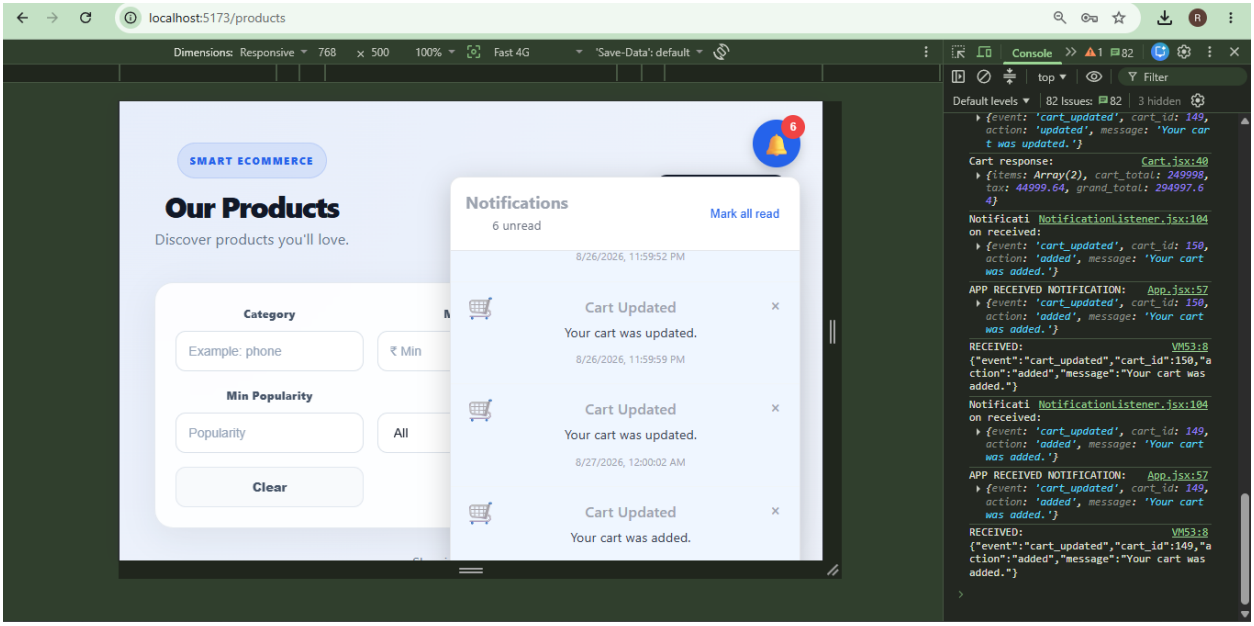


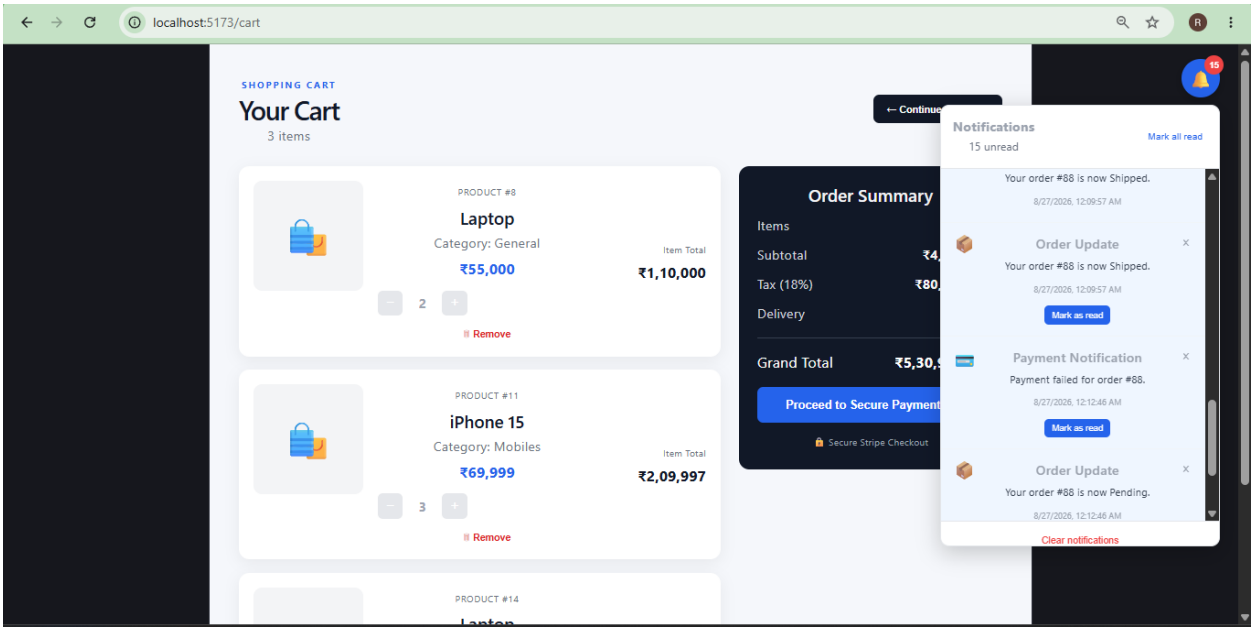
4.1 Fig for Notification API Testing and Swagger Documentation

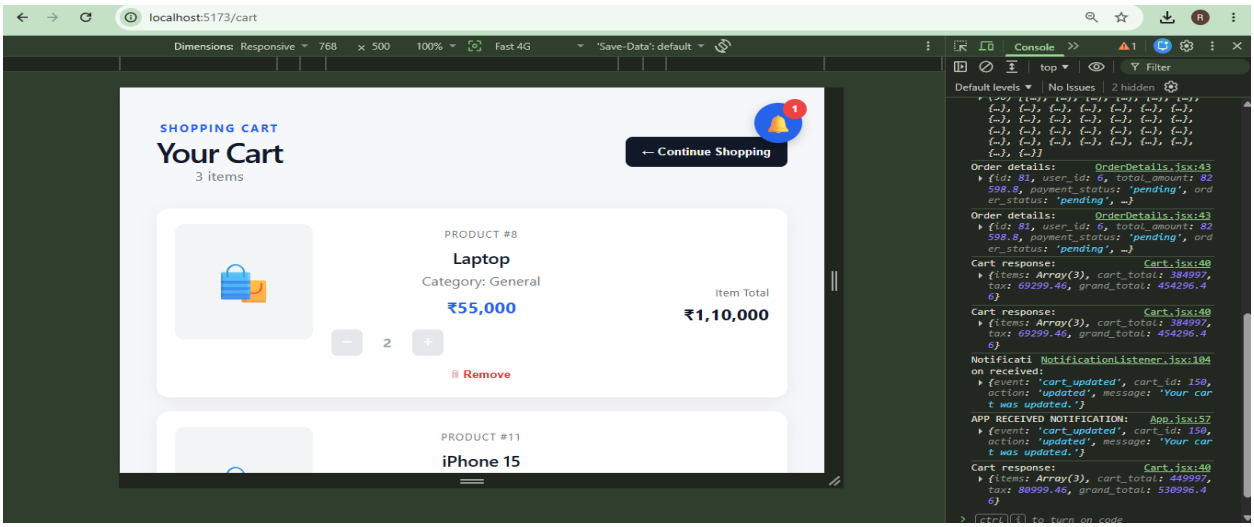


5. Frontend Integration

The React frontend has been integrated successfully with the Notification and WebSocket services.







Notification Page Features

- Notification listing
- Real-time updates
- Read/Unread indication
- Notification timestamp display
- Auto refresh through WebSocket events

Real-Time Features

- Instant notification display
- Live order updates

Final Completion Statement

The assigned **Notification System, Email Integration, and Real-Time Updates** module has been successfully completed.All requested functionalities including notification management, email notifications, WebSocket-based real-time updates, notification APIs, frontend integration, database integration, and API testing have been implemented and verified successfully.The Smart E-Commerce Platform now supports real-time communication, event-based notifications, email delivery, and live user updates, providing an improved user experience and production-oriented architecture.